Project Title

AUTOMATED NETWORK REQUEST MANAGEMENT SYSTEM

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PROJECT OVERVIEW

The **Automated Network Request Management in ServiceNow** project focuses on simplifying and accelerating the handling of network-related service requests. The solution uses ServiceNow's **Service Catalog, custom tables, UI Policies, Catalog Client Scripts, Flow Designer, approval actions, and email notifications** to automate the request management process.

End users can submit network requests through a self-service catalog form. The system captures the required requester and network-related information, validates the submitted data, and stores the request details in a structured custom table. After submission, the request is processed through an automated approval flow. Based on the approval result, the request record is updated and appropriate email notifications are generated.

By reducing manual processing and providing a structured request and approval workflow, the system improves transparency, consistency, and efficiency in network request management.

TABLE OF CONTENTS

1. Introduction
2. Project Objectives
3. Key Features
4. ServiceNow Developer Setup
5. Project Implementation in ServiceNow
6. Field Validation and Automation
7. Flow Designer and Approval Workflow
8. Data Handling and Custom Table
9. Access Control and Security
10. Testing and QA
11. Troubleshooting
12. Screenshots of Output
13. Project Demo
14. Future Enhancements
15. Conclusion

1. INTRODUCTION

Modern enterprises rely heavily on reliable network services to support their day-to-day operations. As organizations grow, the number of network-related service requests also

increases. These requests may include connectivity requirements, existing network service changes, and other network support activities.

Traditional manual request handling can result in delays, incomplete information, repeated communication, and limited visibility into request status. To address these challenges, the **Automated Network Request Management System** provides a structured and automated approach using the ServiceNow platform.

The system allows end users to submit requests through a ServiceNow Service Catalog form. The submitted information is captured through catalog variables and stored in a custom **Database Tables** table. ServiceNow **Flow Designer** is then used to automate record creation, approval processing, status updates, and email notifications.

The solution provides a complete request lifecycle from **request submission to approval and notification**, while reducing repetitive administrative activities and improving transparency for both requesters and IT teams.

2. PROJECT OBJECTIVES

The primary objective of this project is to design and implement an automated solution for managing network-related service requests within ServiceNow.

The major objectives are:

- To provide a user-friendly ServiceNow Service Catalog for submitting network requests.
- To capture requester and network-related information through structured catalog variables.
- To reduce incomplete submissions through appropriate field validation.
- To automatically populate relevant requester information where applicable.
- To store submitted request information in a structured custom table.
- To automate the approval process using ServiceNow Flow Designer.
- To update request records according to the approval outcome.
- To send automated email notifications during the request lifecycle.
- To provide better visibility and traceability of requests.
- To implement appropriate access controls for the custom application data.
- To test the complete request, approval, and notification lifecycle.

3. KEY FEATURES

The major features implemented in the project include:

3.1 Service Catalog

A custom **Network Request** catalog item is provided through the ServiceNow Service Portal. Users can submit their network-related requirements through a structured form.

3.2 Requester Information

The **Requester Information** variable set captures relevant requester details such as user information, email, and phone number.

3.3 Dynamic Form Behaviour

UI Policies and Catalog Client Scripts are used where required to control form behaviour and dynamically handle information displayed or populated on the catalog form.

3.4 Field Validation

Important fields can be configured as mandatory to reduce incomplete request submissions and maintain data quality.

3.5 Custom Database Table

Submitted catalog information is mapped to the custom **Database Tables** table (u_database_tables) for structured storage and further processing.

3.6 Automated Flow

ServiceNow **Flow Designer** automates the request lifecycle. The flow retrieves catalog variables, creates a custom table record, sends notifications, requests approval, evaluates the approval result, and updates the request accordingly.

3.7 Approval Workflow

The system includes an approval stage before the request can proceed to the next stage of processing. The approval outcome is evaluated by the flow.

3.8 Email Notifications

Automated email notifications are used to communicate important request events, such as request creation and approval-related updates.

3.9 Access Control

Access Control Lists (ACLs) are configured for the custom table to control operations such as create, read, write, and delete according to the configured roles.

3.10 Request Tracking

The custom table stores request information and approval status, allowing administrators and authorized users to track the progress of submitted requests.

4. SERVICENOW DEVELOPER SETUP

The project was developed and tested using a **ServiceNow Personal Developer Instance (PDI)**.

The development environment was used to configure:

- Service Catalog
- Catalog Variables

- Variable Sets
- UI Policies
- Catalog Client Scripts
- Custom Tables
- Access Control Lists
- Flow Designer
- Approval Actions
- Email Notifications
- Service Portal

The project was implemented within the ServiceNow development environment and tested using sample request submissions.

5. PROJECT IMPLEMENTATION IN SERVICENOW

The implementation was carried out in multiple stages:

Phase 1 – Catalog Creation

A Service Catalog item named **Network Request** was created to allow users to submit network-related requests.

Phase 2 – Form Configuration

Catalog variables were configured to collect the required request information, including:

- Requested For
- Mobile Number
- Type of Connection
- Existing ID
- Total Amount
- Mode of Payment
- Address

A **Requester Information** variable set was also configured for requester-related information.

Phase 3 – Custom Table

A custom table named **Database Tables** was created to store the submitted request information.

The table contains fields for important request information such as:

- Database Number

- Requested For
- Mobile Number
- Type of Connection
- Existing ID
- Total Amount
- Mode of Payment
- Address
- Approval Status
- Assigned To

Phase 4 – Flow Designer

A Flow Designer flow named **Network Request** was configured to automate the request lifecycle.

The flow performs the following major operations:

Service Catalog Request → Get Catalog Variables → Create Database Tables Record → Send Email → Ask for Approval → Check Approval Result → Update Record

Phase 5 – Testing

The complete workflow was tested using sample catalog submissions. Request creation, record generation, approval processing, email notification, and request status updates were verified.

IMPLEMENTATION

Phase:2

Step:1.1 Table Creation

Navigate to: System Definition > **Tables**.

Click **New** to create a new table.

Fill **in Table Information**:

Label: Database Tables

Name: u_database_tables

Click **Submit** to create the table.

Step:1.2 Create Fields in Database Table

Navigate to **All** → **System Definition** → **Tables**.

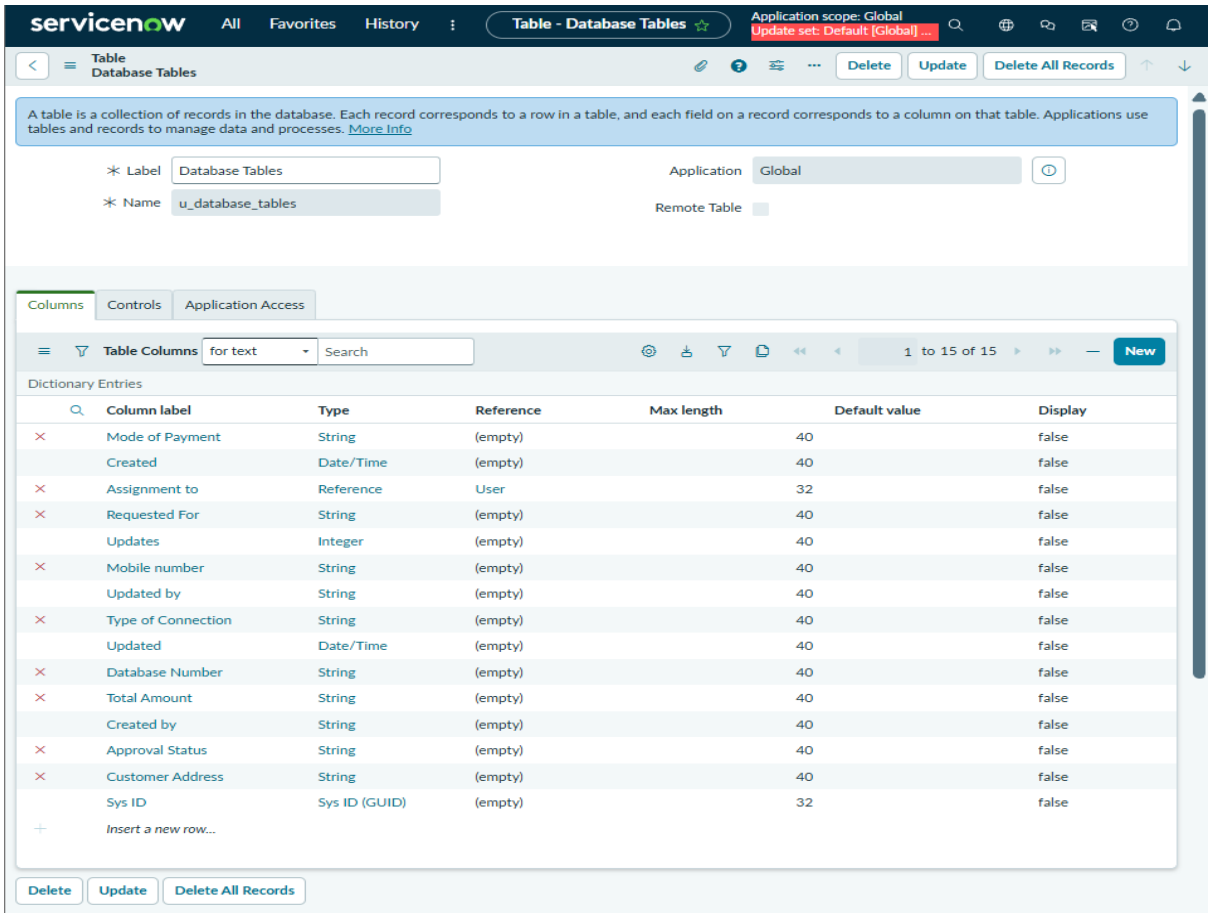
Search for **Database Tables** in the table list.

Open the **Database Tables** table.

Create fields using **Insert new record** at table columns

Create required fields which are shown in the img

Verify the newly created fields by opening a **Database tables** records.



Column label	Type	Reference	Max length	Default value	Display
Mode of Payment	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Assignment to	Reference	User	32		false
Requested For	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Mobile number	String	(empty)	40		false
Updated by	String	(empty)	40		false
Type of Connection	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Database Number	String	(empty)	40		false
Total Amount	String	(empty)	40		false
Created by	String	(empty)	40		false
Approval Status	String	(empty)	40		false
Customer Address	String	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false

Step: 2.1 Creation of Approval Request on Network Database Table

Activity 1: Creation of Related List

Navigate to **System Definition > Relationships**.

Click **New** to create a new relationship.

Fill in the following details:

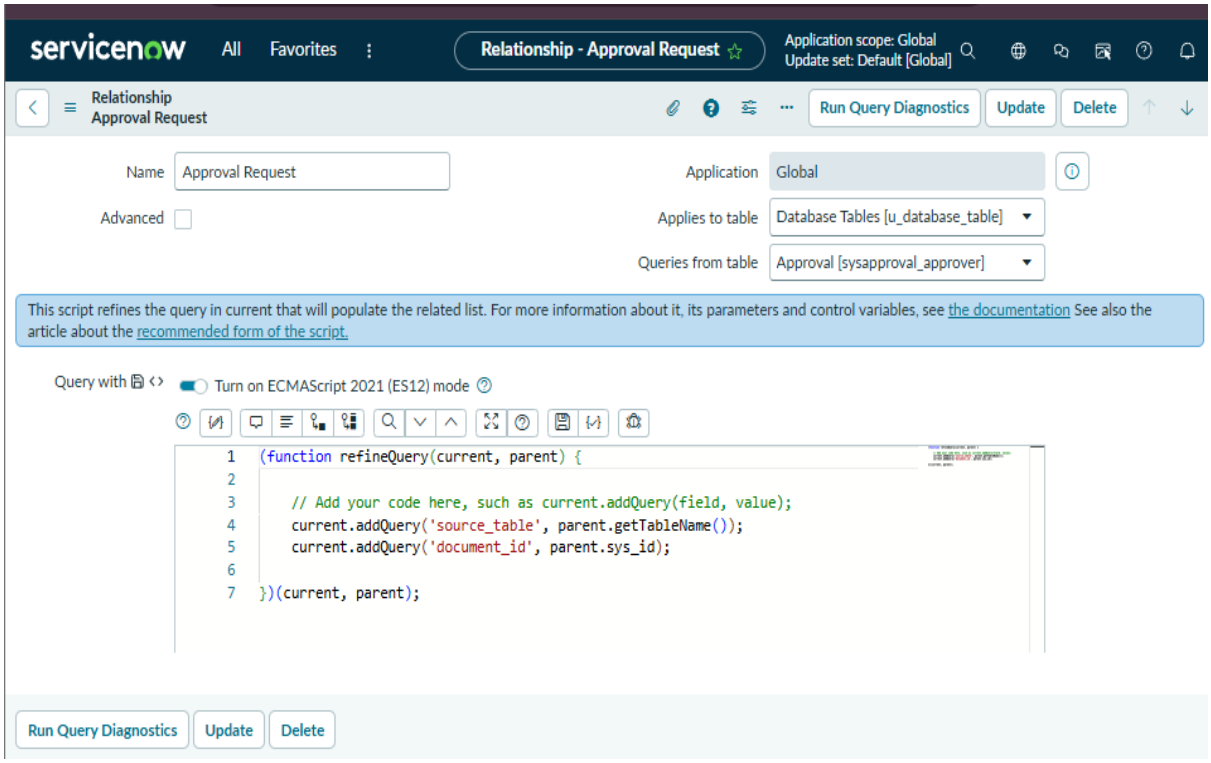
Name: Approval Request

Applies to Table : Database table

Queries from Table : sysapprovals table.

Active: Make sure it's set to **True**.

Save the relationship.



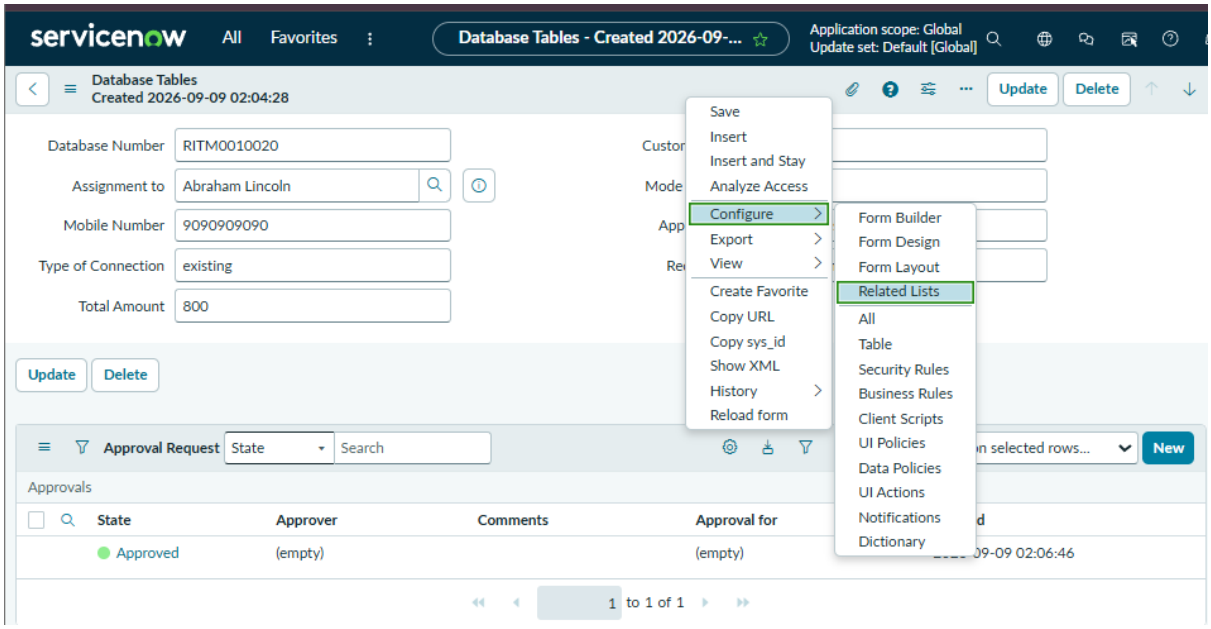
Activity 2: Adding Related List to the Table

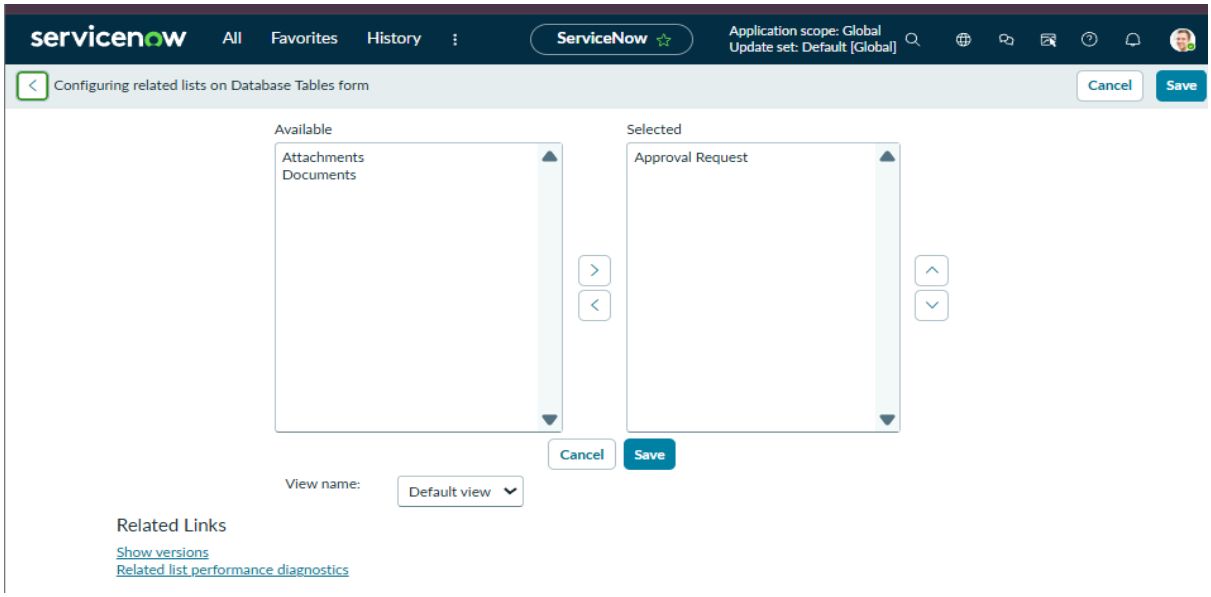
Through tables->Database tables -> in related links **click Show List -> create new** ->fill the fields as shown in fig .

Now open that just now created record.

Add a **Related List** widget to the form.

Select the **Related List** you want to show





Phase-3

Interface Design

Activity 1: Creation of Service Catalog

medium

Description

Navigate to Application navigator

Click on All >> search for Service Catalog

Under Service Catalog>> Maintain items

Click on New

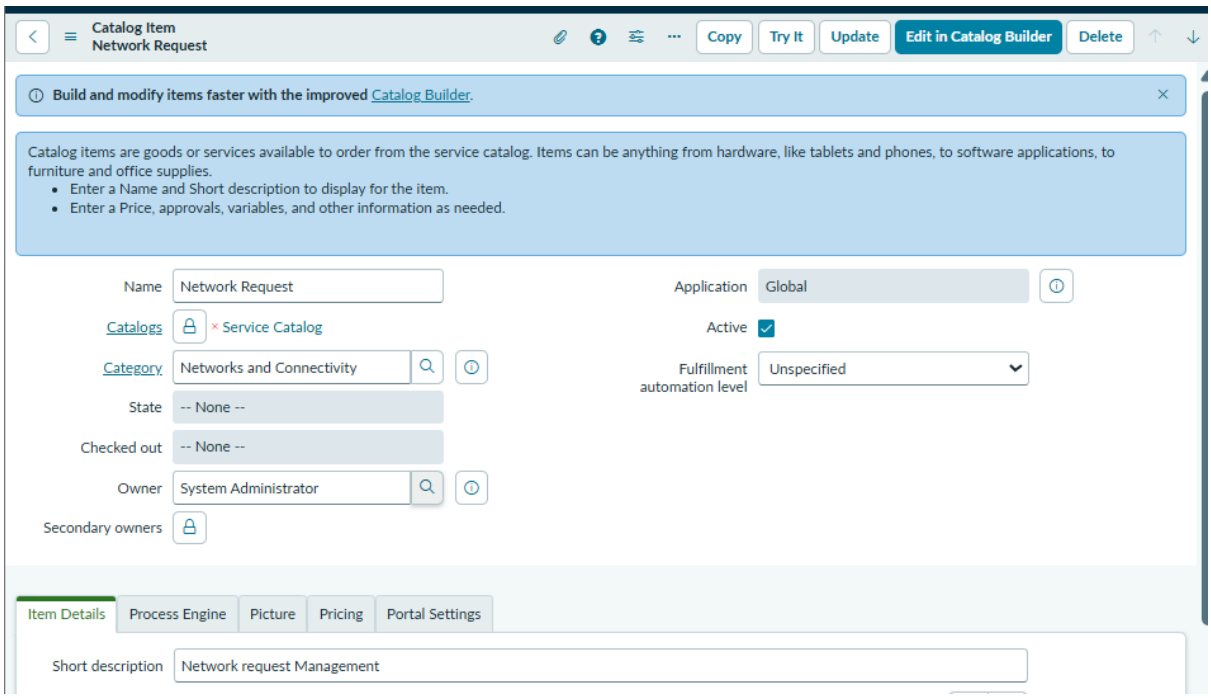
Fill the details >> Name– Network Request

Select Catalog>> Service Catalog

Select Category>> Networks and Connectivity / Network Standard changes

Fill the Short Description as Network request Management

Click on Save.



Open the catalog item just created.
Scroll down to the **Variables** related list and click **New** to create form fields.

Select Variables type as Single, Multi line text, reference, choices etc as per requirement

Catalog item– Network Request

Order–100,200,300,,,,

Question– provide the variable label

Name–provide the variables name(used for scripting)

Tooltip– this will appear when cursor overed on the field

Example text – this will suggest what we need to enter on the field.

Mandatory, Read-Only– need to configure on demand

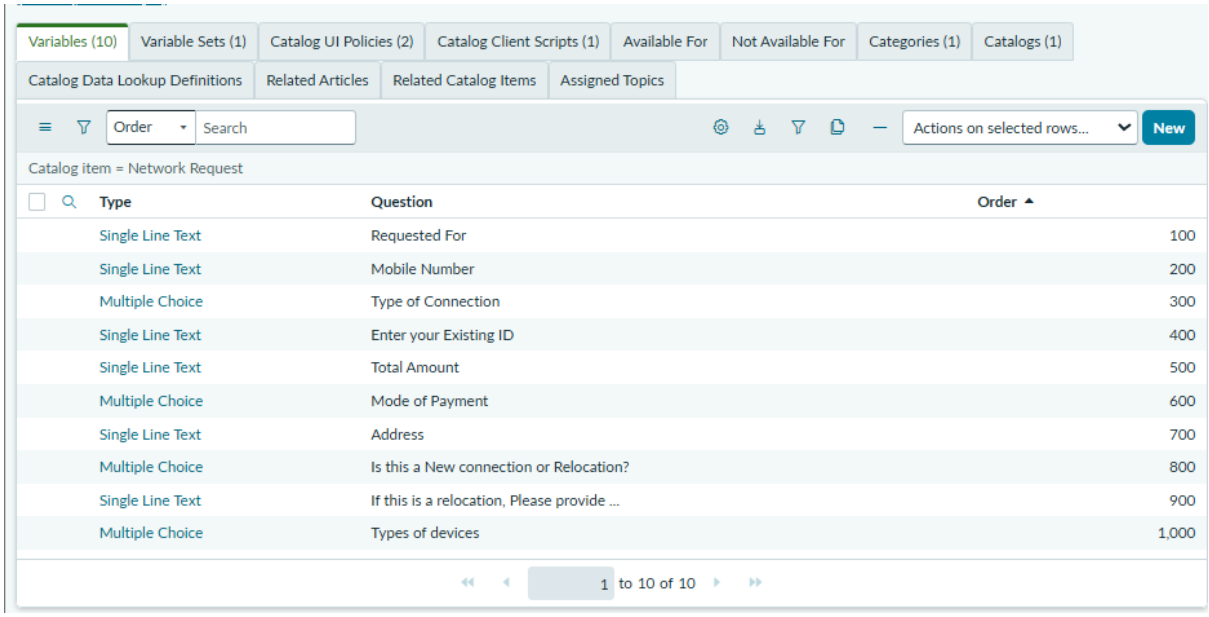
Auto populate– need to select dependent variable, apply dot walking to get selected value.

Click on Save or Submit.

I created 10 variables 7 are enough

Note : For multiple choices ,Default was new ,UPI ,Laptop etc based on fields

Don’t create choices which are add in to default tab.



Activity 4: Variable Set Configuration

Description

To enhance form usability create a Variable set

Navigate to the **Variable Sets** with in the Catalog Item.

Follow the same procedure as we used for Variables Creation, for the variable set as well.

Apply variable sets to the catalog item.

Variables Details

S.NO	Variable Name	Type	Reference / Configuration
1	**Opened on behalf of**	Reference	Reference → **User [sys_user]** table
2	**Email ID**	Single Line Text	Auto-populate from **Opened on behalf of**
3	**User Name**	Single Line Text	Auto-populate from **Opened on behalf of**
4	**Phone Number**	Single Line Text	Auto-populate from **Opened on behalf of**

Variable Set
Requester Information

Update | Delete

* Title	Requester Information	Application	Global
* Internal name	requester_information	Display title	☑
Order	100	Layout	2 Columns Wide, one side, then the (
Type	Single Row		
Description			

Update | Delete

[SN Utils] Versions (13)

Variables (4) | Catalog UI Policies | Catalog Client Scripts | Included In (1) | Catalog Data Lookup Definitions

Order | Search | Actions on selected rows... | New

Variable set = Requester Information

Name	Type	Question	Order
opened_on_behalf_of	Reference	Opened on behalf of	100
email_id	Single Line Text	Email ID	200
user_name	Single Line Text	User Name	300
phone_number	Single Line Text	Phone Number	400

1 to 4 of 4

NOTE : Follow the same procedure to create all four variables and for the auto populate the vales follow the below pictures

Variable
Email ID

Copy | Update | Delete

Application	Global	Active	☑
Type	Single Line Text	Mandatory	☐
Order	200	Read only	☑
Variable set	Requester Information	Hidden	☐
		Disable automatic slot fill based on user context	☐

Question | Annotation | Type Specifications | Default Value | Auto-populate | Permission | Availability

Specify the **Question** that explains the options available to the end user when ordering the item

* Question	Email ID
* Name	email_id
Conversational label	
Tooltip	Email ID of the selected user
Example Text	Automatically populated

Question | Annotation | Type Specifications | Default Value | Auto-populate | Permission | Availability

Dependent question	Opened on behalf of
Reference	User [sys_user]
Dot walk path	Email

Note: Follow the same procedure to auto populate values for all other variables

Step :3.3 UI Policy

Procedure

Navigate to **Maintain Items**

Open **Network Request** item

In related list, we have Catalog UI policy

Click on New button to configure New UI policy

Select Applies to as Catalog item

Select catalog item as Network Request

Provide short description, if required

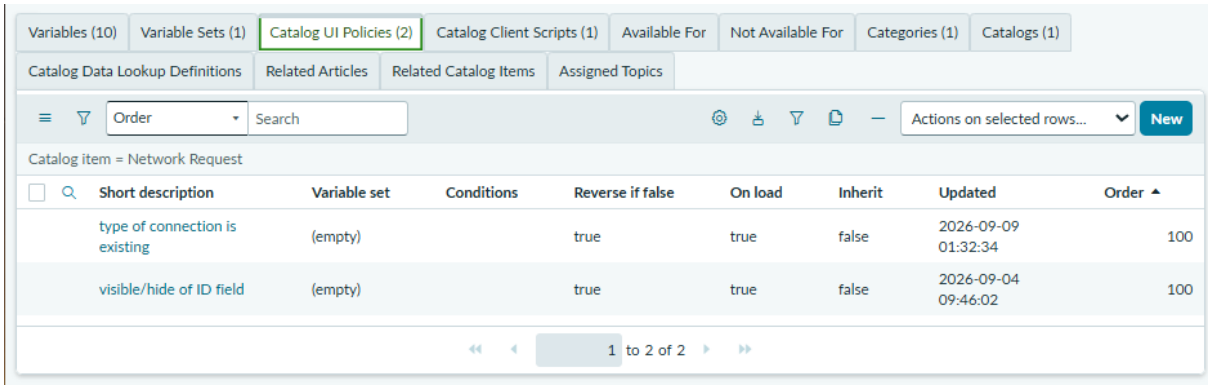
Apply condition>> **types of Connection** is **Existing**

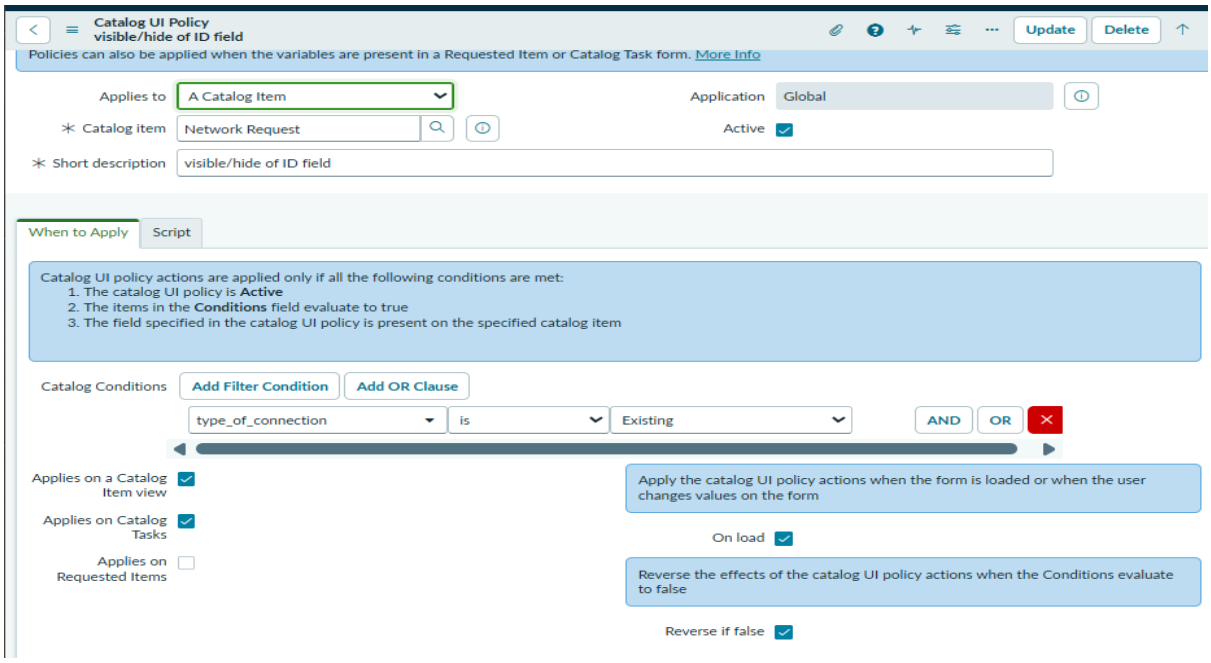
Click on save, after saving the form will get UI policy actions in the related list

Click on New button to configure new UI Policy action, and select the variable which we want to display on condition

Make Visible True as per our requirement

Update the UI Policy and Test the same on Catalog form.





Step:3.4 **Automation**

Activity 1. Creation of Flow

Navigate to Flow designer home page

Click on New to create a new flow

Provide flow name as **Network Request**

Provide description of flow

Click on Build flow.

Activity 2. Configuring Trigger

Click on (+) Icon to Configure the Trigger

Select Trigger as Application >> Service catalog

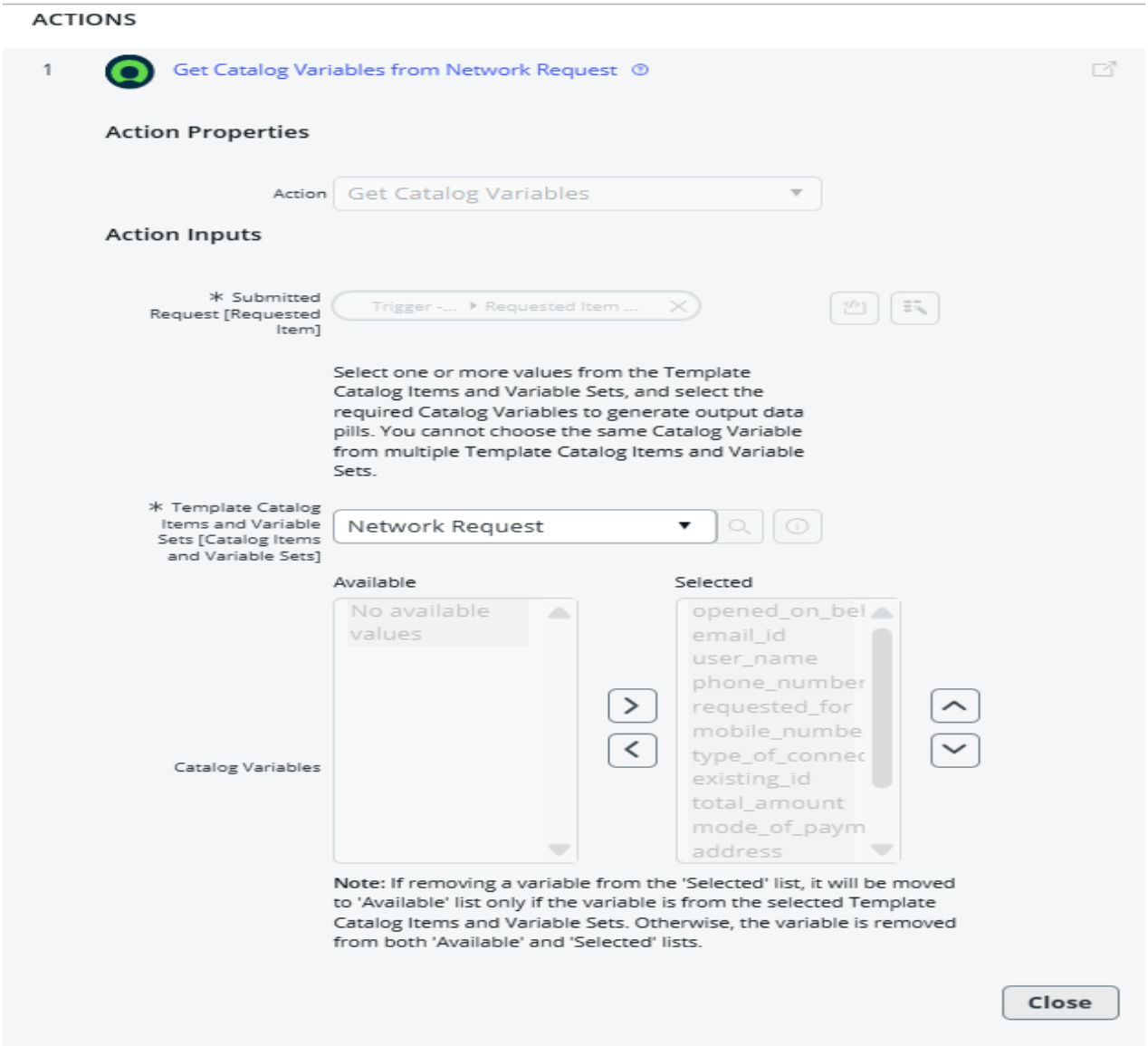
Click on Done.

Activity 3. Configuring Actions

Click on Actions button to configure new action

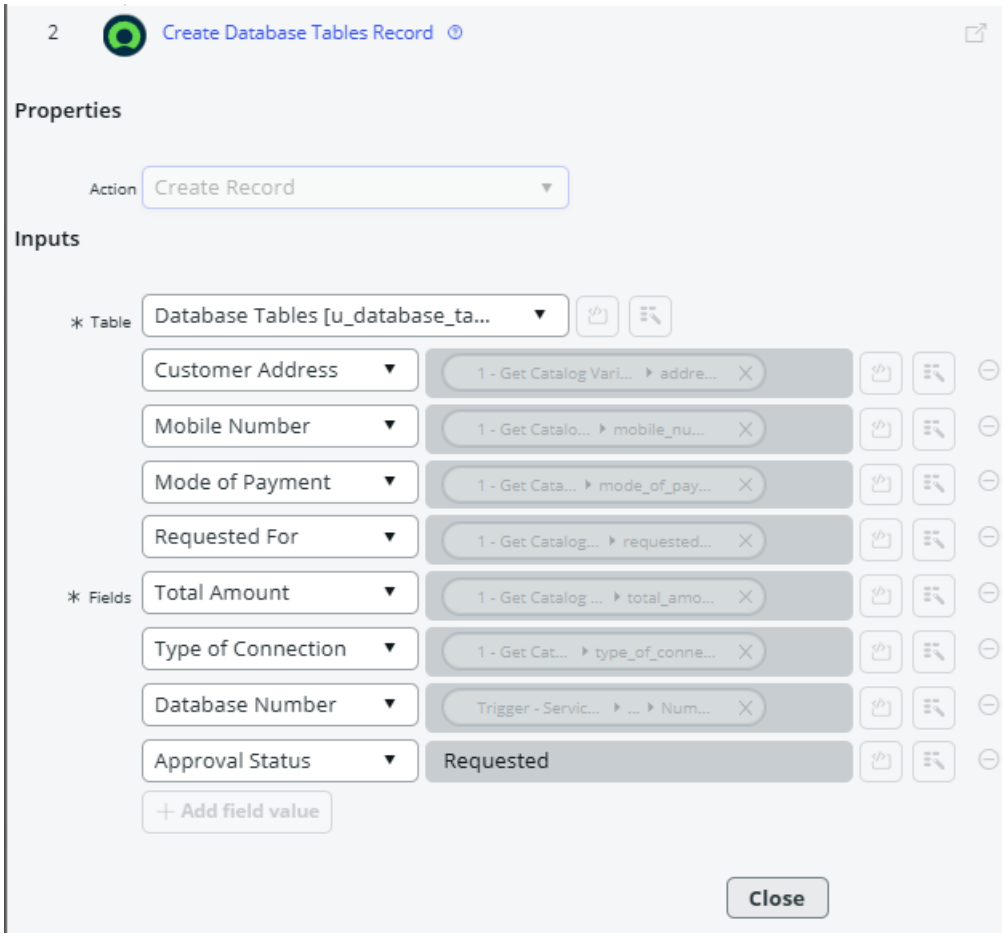
Get Catalog Variables

- Click on Action, search for Get Catalog Variables
- Select Get Catalog Variables
- Action Inputs>> Trigger>>service catalog>>Requested Item
- Template catalog items >> Select table >> Network Request
- Select the Required Variables and Move to the selected area.
- Click on done



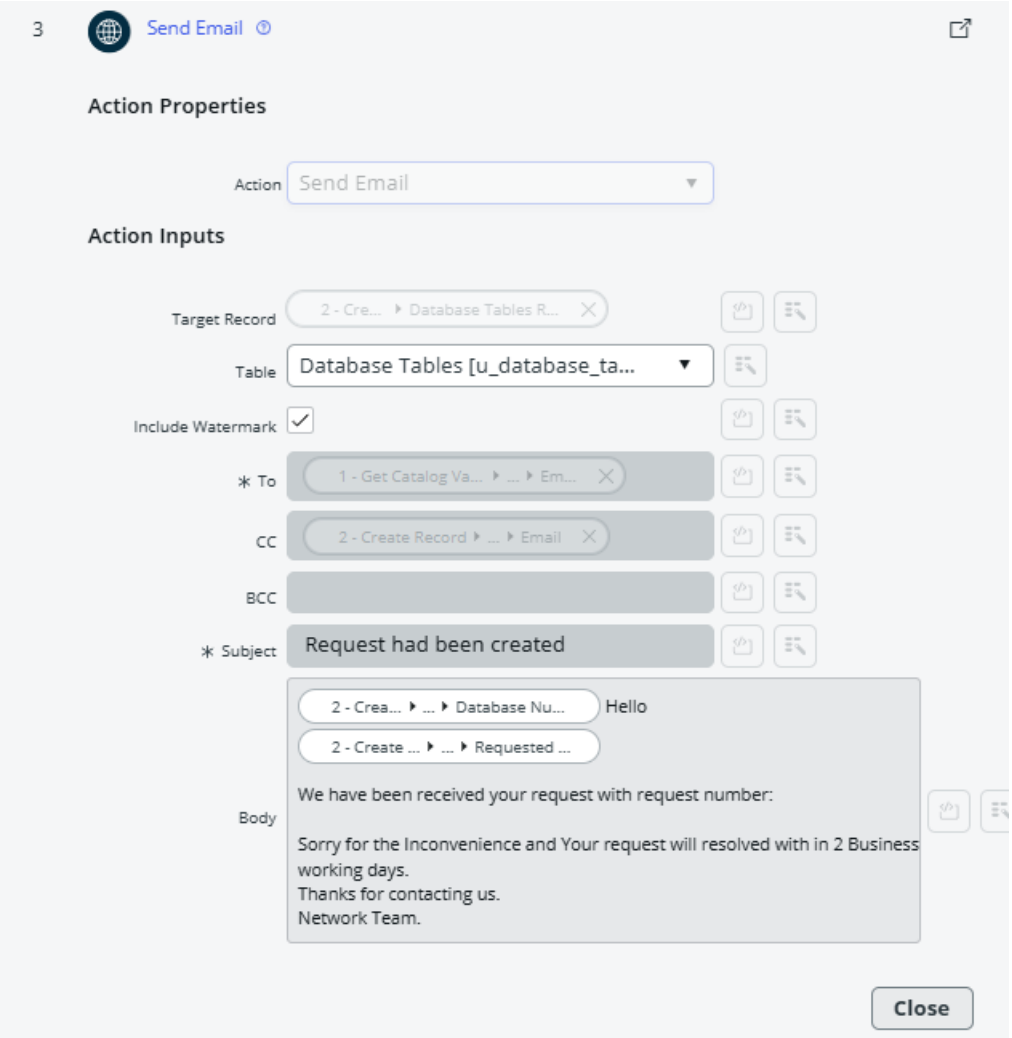
Create Record

- Select action as Create Record
- Select table as **Database tables**
- Click on Add fields button to configure the fields
- Configure the Required fields as shown in the below picture
- Click on done



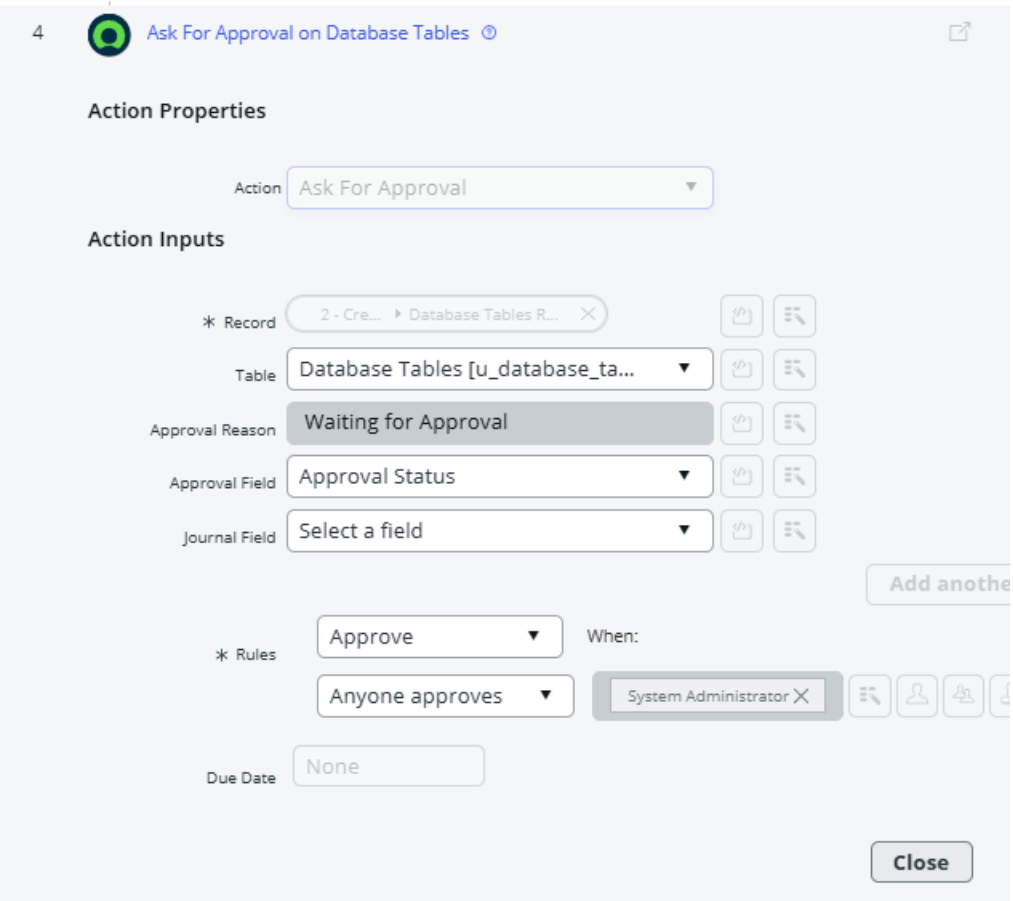
Send Email

- Select action as Send Email
- Select target record >> Create record>> database tables
- Table will be selected automatically
- Configure To, CC, BCC as per our requirements (select static/dynamic)
- Provide Subject & Body as shown in the below picture
- Click on done



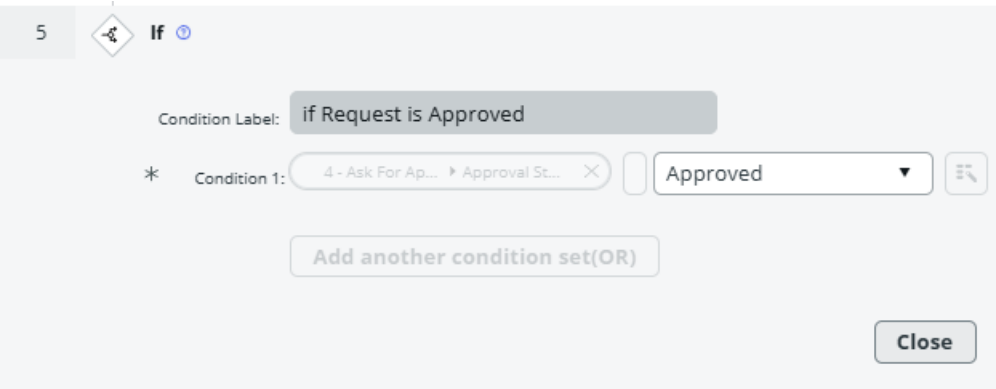
Ask for approvals

- Select action as Ask for Approval
- Select target record >> Create record>> database table
- Provide Approval Reason>> Waiting for approval
- Configure approval rules>> Approve, reject, approve/reject
- Select approvals as Anyone approves, everyone approves etc.
- We can select approvals like static/dynamic as shown below
- Click on done



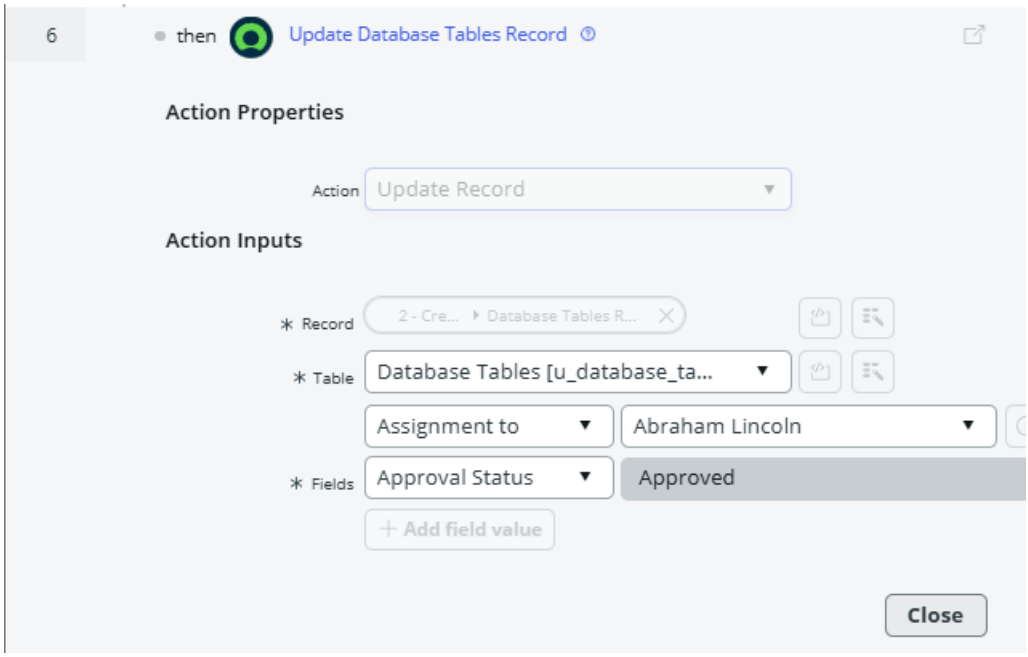
Flow Logic

- Select action as flow logic and Select If condition
- Apply condition >> Ask for approvals state is Approved/Rejected as per requirement
- Click on done



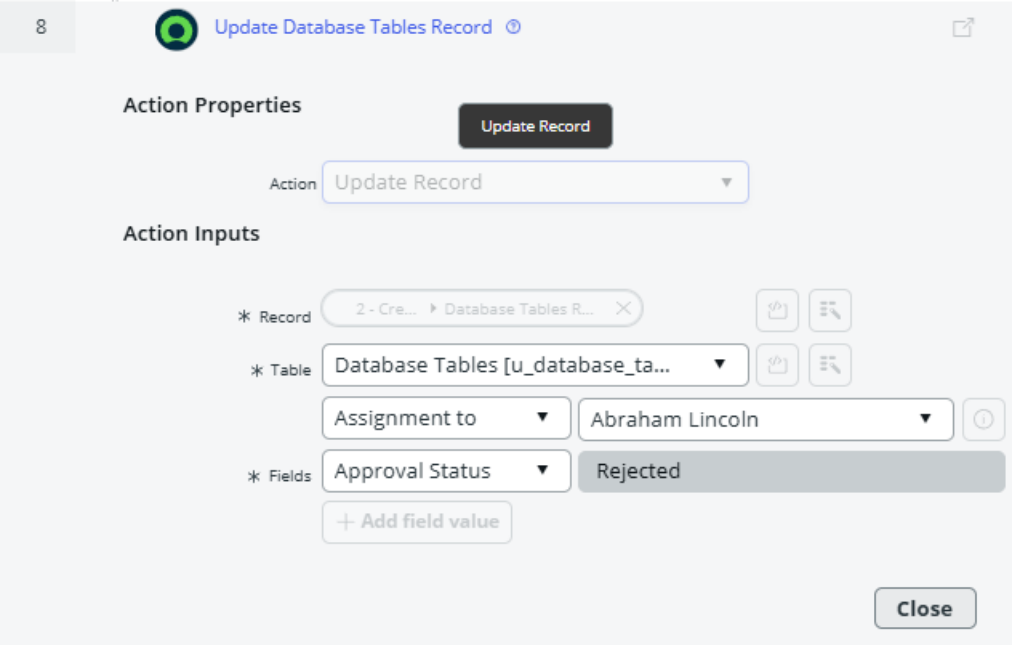
Update Record

- Select action as Update Record
- Select record as >> create record>> database tables
- Table will be selected automatically
- Configure the fields as per requirement, as shown in below
- Click on done



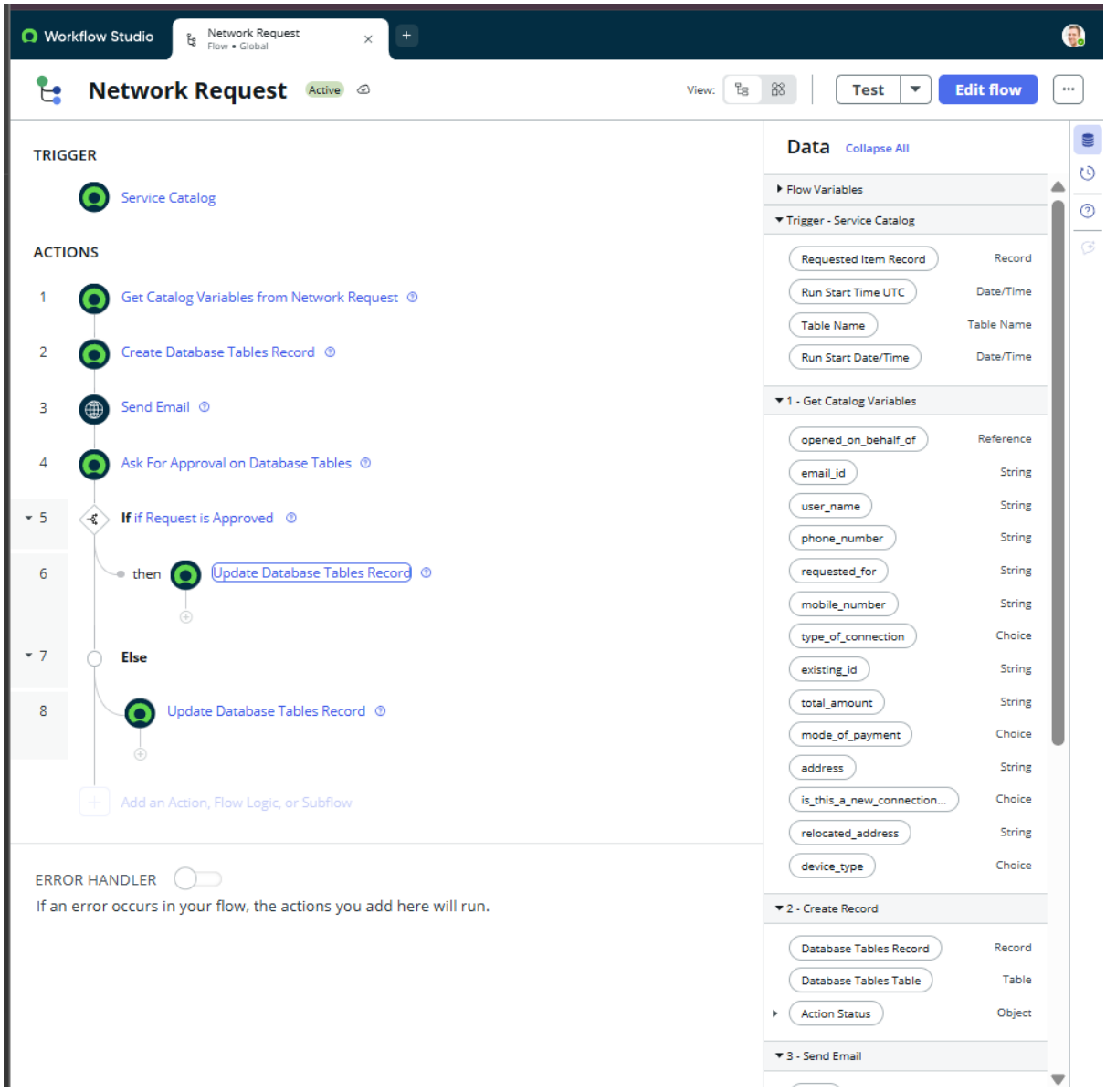
Flow Logic(Else)

- Select action as flow logic and Select Else condition
- In this Else condition select Update record as shown below



Note: Configure the actions and Logics as shown in the flow chart.

Full Flow Chart



ACTIVATE it .

Then the page appear like this

Step :3.2 Navigation Flow

Procedure:

Login to ServiceNow PDI

Copy the Instance domain ex: https://dev190678.service-now.com.

Paste the URL in the Next tab and add Prefix SP to the URL. ex: https://dev190678.service-now.com/sp.

Search for **Network Requests.**

Fill the required details and click on submit

New Requests will be generated with request numbers and users will get particular emails on the same.

servicenow

Knowledge Catalog Requests System Status

Home > Search

network request

Sources

All
Knowledge Bases
Catalogs

All results for "network request"

Network Request
Network request Management

servicenow

Knowledge Catalog Requests System Status Cart Tours

Home > Service Catalog > Standard Changes > Network Standard Changes >
Networks and Connectivity > Network Request

Search Catalog

Network Request

Network request Management

* Indicates required

Requester Information

*Opened on behalf of: Abraham Lincoln

User Name: abraham.lincoln

Email ID: abraham.lincoln@example.com

Phone Number: (555) 555-0004

*Requested For: Abraham Lincoln

*Mobile Number: 9638527412

*Type of Connection
- Existing
- New
- New

*Total Amount: Enter total amount

*Mode of Payment
- UPI
- Card
- UPI

*Address: Eluru

*Is this a New connection or Relocation?

Quantity: 1

Delivery Time: 2 Days

Add to Cart

Save as Draft

Order Now

Required information

Total Amount

Order Confirmation

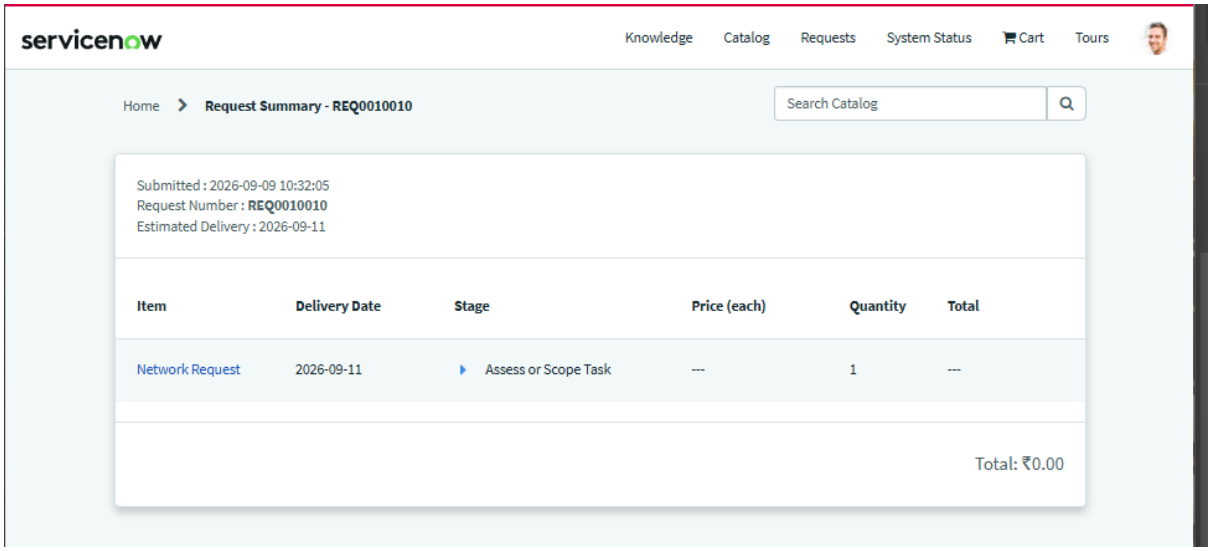
Request for

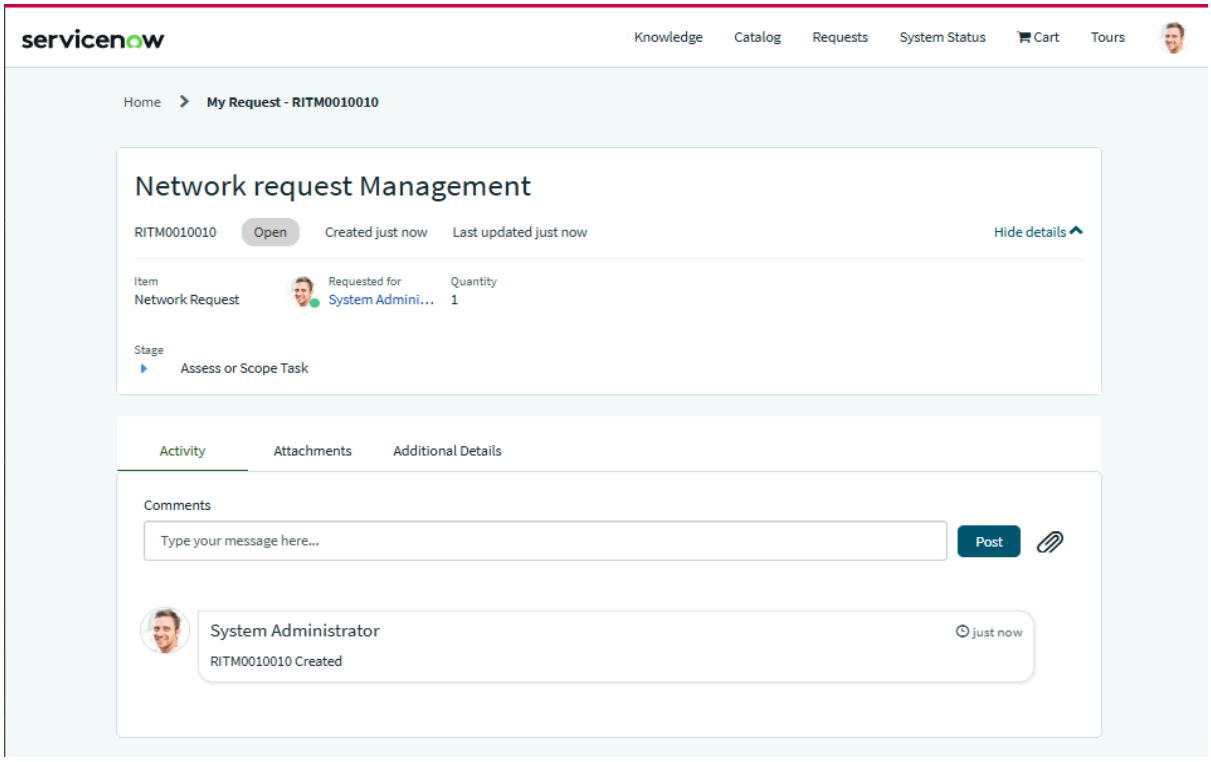
System Administrator

Delivery Information (Optional)

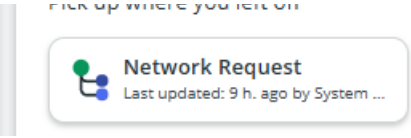
Special instructions (Optional)

Cancel Checkout





Copy the request number ex: RITM0010010



Workflow Studio | Network Request Flow • Global

Network Request

Active

View: | Test | Edit flow

TRIGGER

Service Catalog

ACTIONS

1 Get Catalog Variables from Network Request

2 Create Database Tables Record

3 Send Email

4 Ask For Approval on Database Tables

5 **If** if Request is Approved

6 then Update Database Tables Record

7 **Else**

8 Update Database Tables Record

Add an Action, Flow Logic, or Subflow

Data

Collapse All

Flow Variables

Trigger - Service Catalog

Requested Item Record	Record
Run Start Time UTC	Date/Time
Table Name	Table Name
Run Start Date/Time	Date/Time

1 - Get Catalog Variables

opened_on_behalf_of	Reference
email_id	String
user_name	String
phone_number	String
requested_for	String
mobile_number	String
type_of_connection	Choice
existing_id	String
total_amount	String
mode_of_payment	Choice
address	String
is_this_a_new_connection...	Choice
relocated_address	String

Test flow

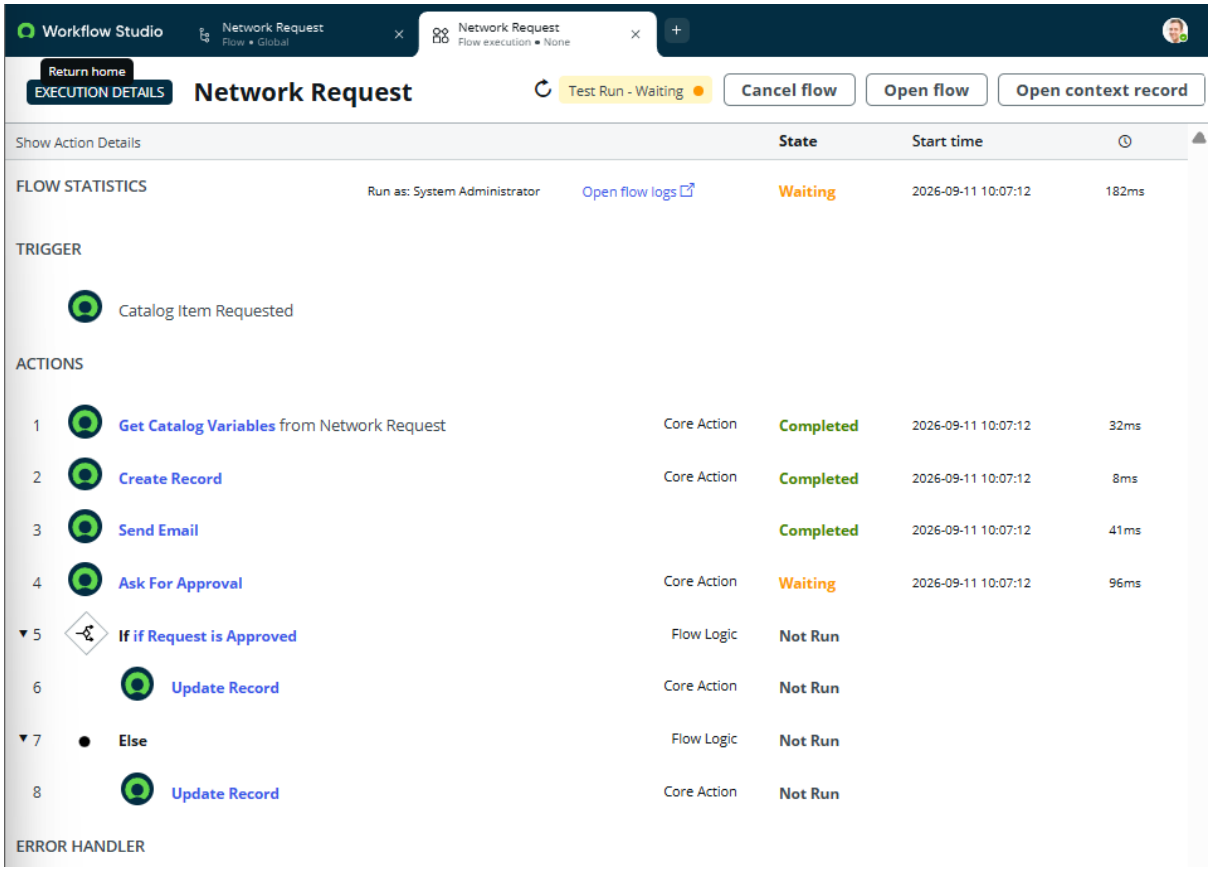
Run your flow to make sure it has no errors before you activate it. When the test finishes running, check the execution details to see each action's configuration, runtime values, and the log messages for any errors that occurred.

* Requested Item Record: RITM0010010

Your test has finished running. View the flow execution details.

☐ Run test in background

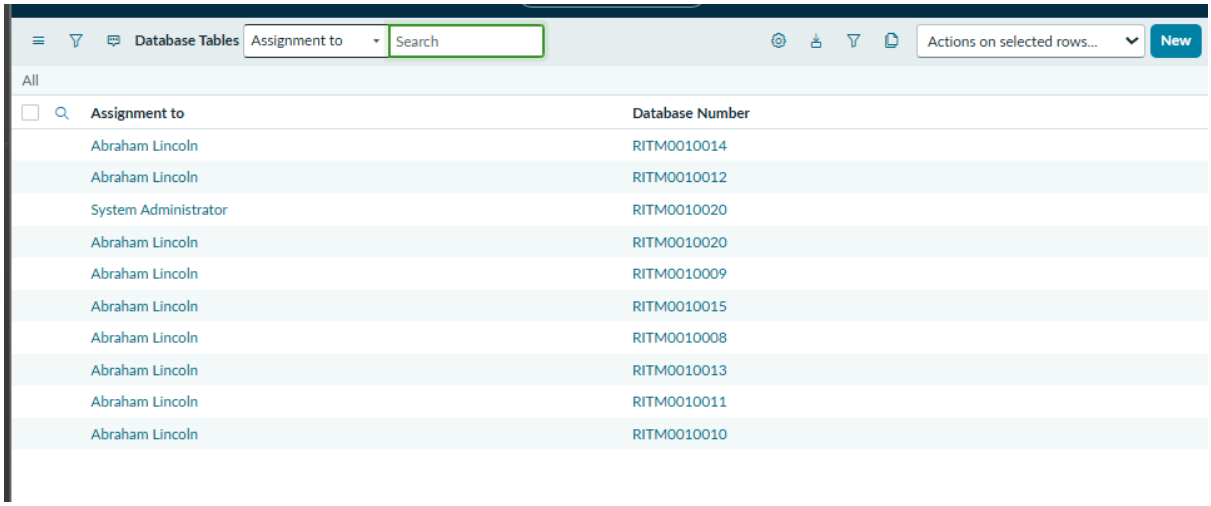
Cancel | Run Test



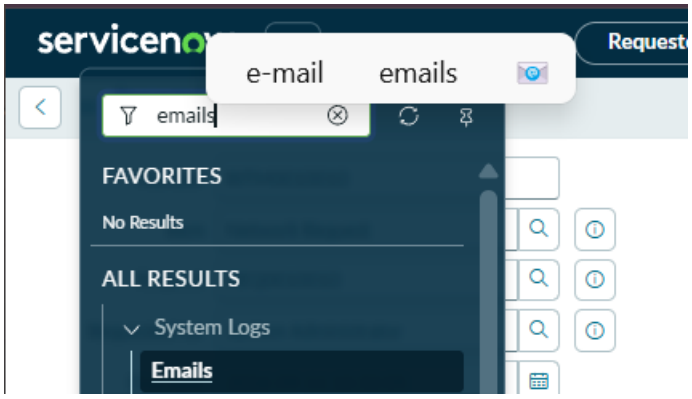
Check those are really work or not?

Goto all >> database tables

Look for RITM0010010 then create record is done.



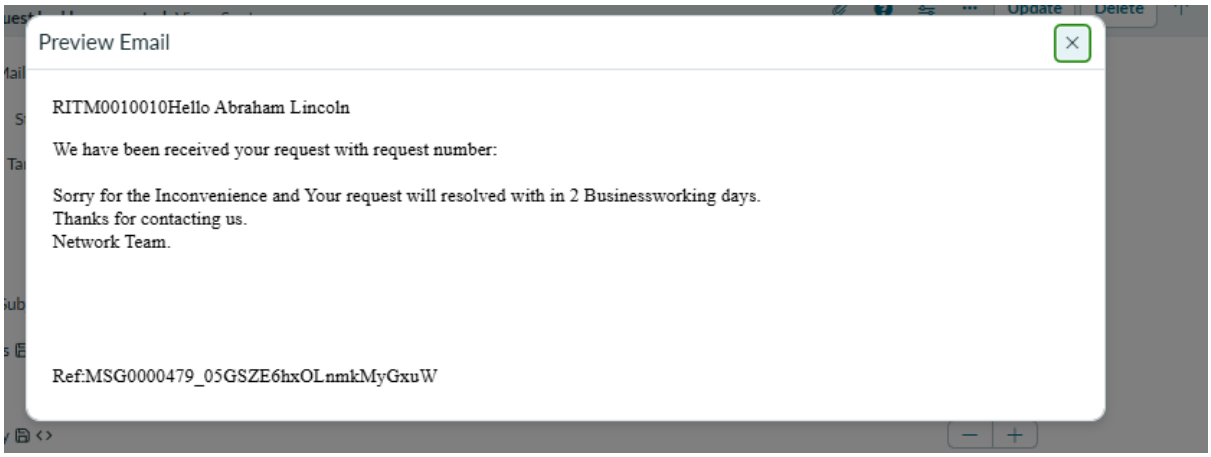
Now goto emails under system logs



Now open **Request had been created** email

Preview email

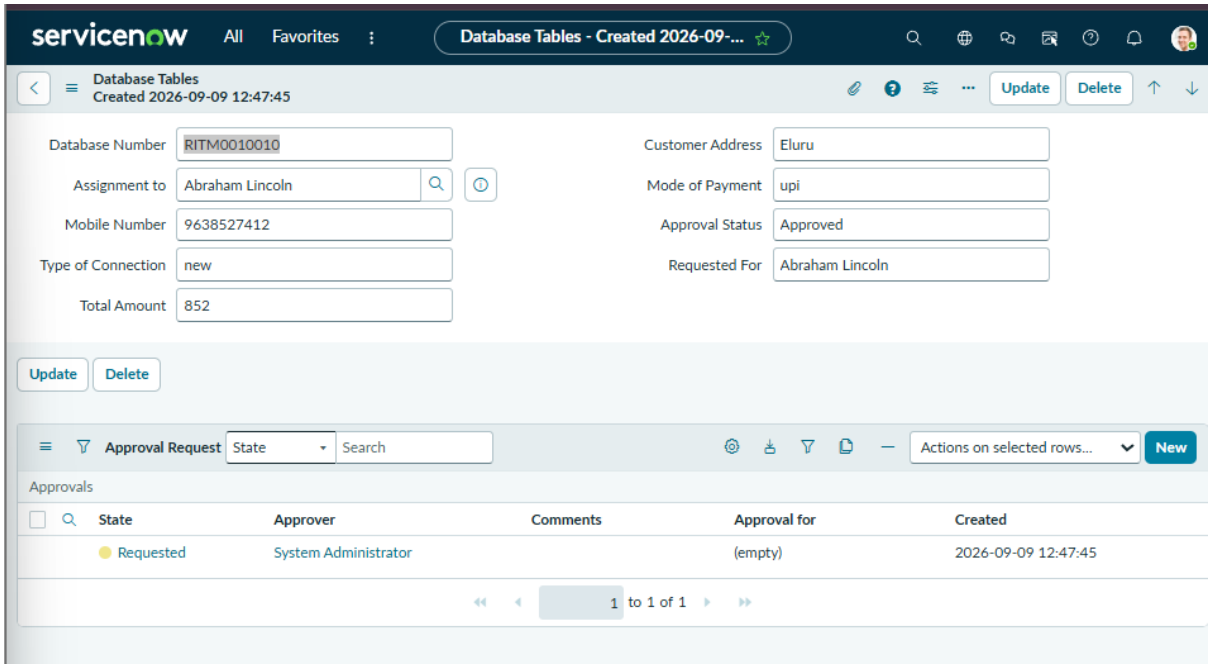
done



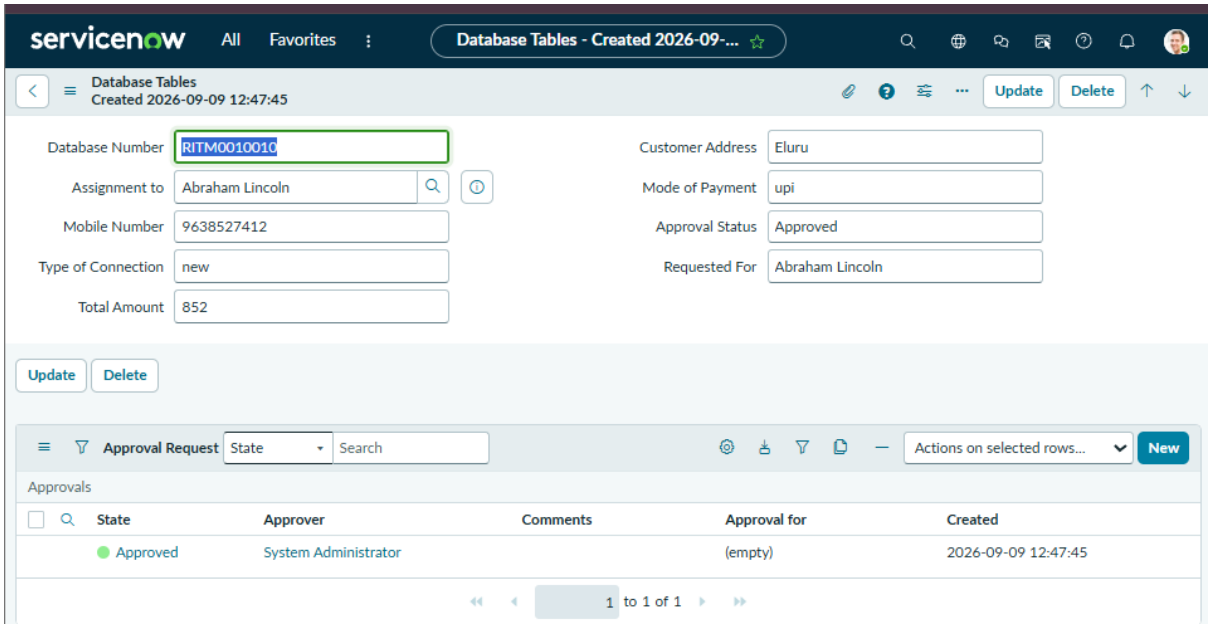
Goto all >> database tables

Look for RITM0010010

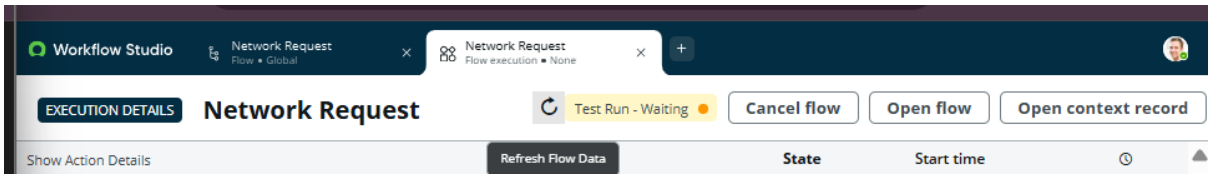
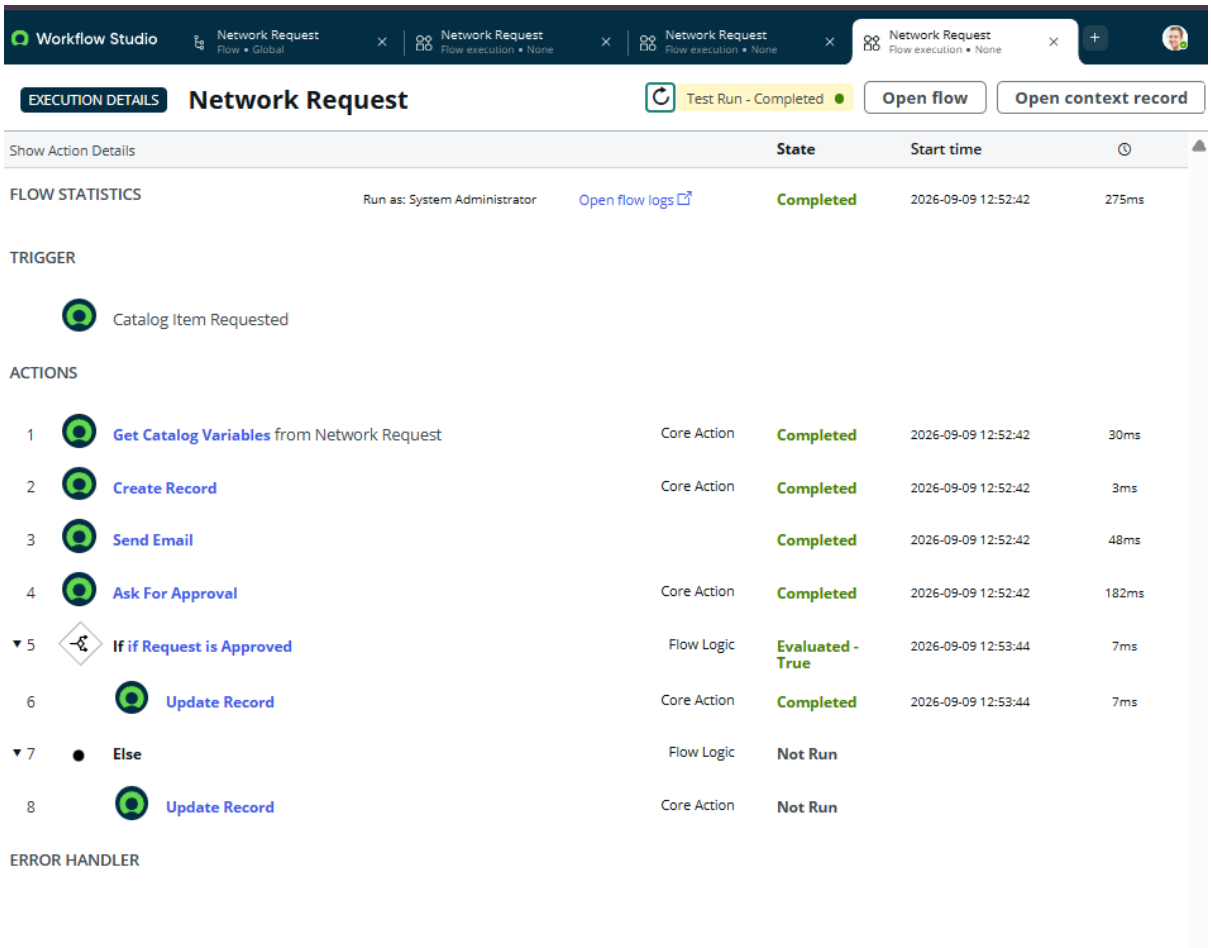
open that record



Click requested update it to approve



Later do this



Test run